

FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

Six benches, already wired and running

- Ten minutes each. Hard timer
- Change one parameter, answer one question
- You wire nothing today

Then build one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The rotation is the spine of the session, so keep it moving.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	30 min	Bench tour: touch all six, five minutes each
0:40	10 min	Round-the-room: what surprised you
0:50	50 min	Tasks. Grab a board, work at your own pace
1:40	15 min	Blind test on task 5
1:55	5 min	Pack down

THE SIX BENCHES

01 ERM COIN MOTOR

VIBRATION

An off-centre weight on a motor shaft.

The same part as in your phone.

Speed and strength are welded together.

"Gentle but fast" is not on offer.

USED FOR

phone notifications, game controller rumble,
anything worn under clothing.

part 10mm 3V coin ERM

drive 2N2222 + 1N4148

pin D9 (PWM)

draw ~75 mA

spin-up 20-40 ms

cost ~12 kr

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

02 LRA + PIEZO DISC

VIBRATION

Two ways out of bench 01's compromise.

LRA: resonance in 5 ms, stops just as fast.

Piezo: near-instant, almost no power.

A tick, not a thump.

USED FOR

keyboard clicks, Apple Watch taps, anything where lag would feel broken.

parts LRA 10mm + piezo

driver DRV2605L (I2C)

pin A4 SDA / A5 SCL

effects 123 waveforms

cost ~90 kr

CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?

03 SOLENOID TAP

IMPACT

One 15 ms pulse against a fingertip.

A knock reads as a person tapping you.

Buzzing reads as a machine.

CARD QUESTION

How many taps before it goes from alert to nagging?

USED FOR

navigation cues on the body, braille cells,
alerts that must cut through noise.

part 5V push-pull

drive MOSFET + flyback

pin D6

peak ~1.1 A

duty <= 25%, gets hot

cost ~45 kr

04 CAPACITIVE TOUCH

TOUCH INPUT

Bare finger on a surface. Servos react.

No button. No sensor you can point at.

The input is a wire behind a plate.

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

USED FOR

invisible controls in wood or fabric, waterproof panels, touch-sensitive prototypes.

sense ADCTouch on A0

baseline 25-sample roll

trigger 1.01x average

debounce 100 ms

servos D4 + D13

cost ~0 kr, a wire

BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

05 PELTIER WARM / COOL

THERMAL

Heats one side, cools the other.

Flip the current, it reverses.

Sit with it. Seconds before you are sure.

Hot alternating with cold reads as lukewarm.

USED FOR: slow ambient state, not alerts.

"The room is busy", never "you have mail".

part TEC1-12706
drive L298N H-bridge
pin D3 PWM / D4-5
draw 2-4 A, bench PSU
onset 3-8 s
cap 45 C in software

06 THE SAME SIGNAL, TWICE

AUDIO + TOUCH

One driver. 40 Hz you feel, 400 Hz you hear.

Together is not louder. It is more certain.

Two channels, same message.

CARD QUESTION

With ear defenders on, does the signal still work?

USED FOR

noisy or bright environments, and accessibility.
If one channel fails, one lands.

part bone-conduction

amp PAM8403 class-D

pin D11 (tone)

felt ~40 Hz

heard ~400 Hz

cost ~110 kr

NOW MAKE ONE DO SOMETHING

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

- 01 Make something spin**
- 02 Make something tap**
- 03 Make something warm**
- 04 Make it notice you**
- 05 Make it answer back**

1 to 3 are output. 4 is input. 5 is both.

You are not expected to finish all five.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one



INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch. A wire. See bench 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

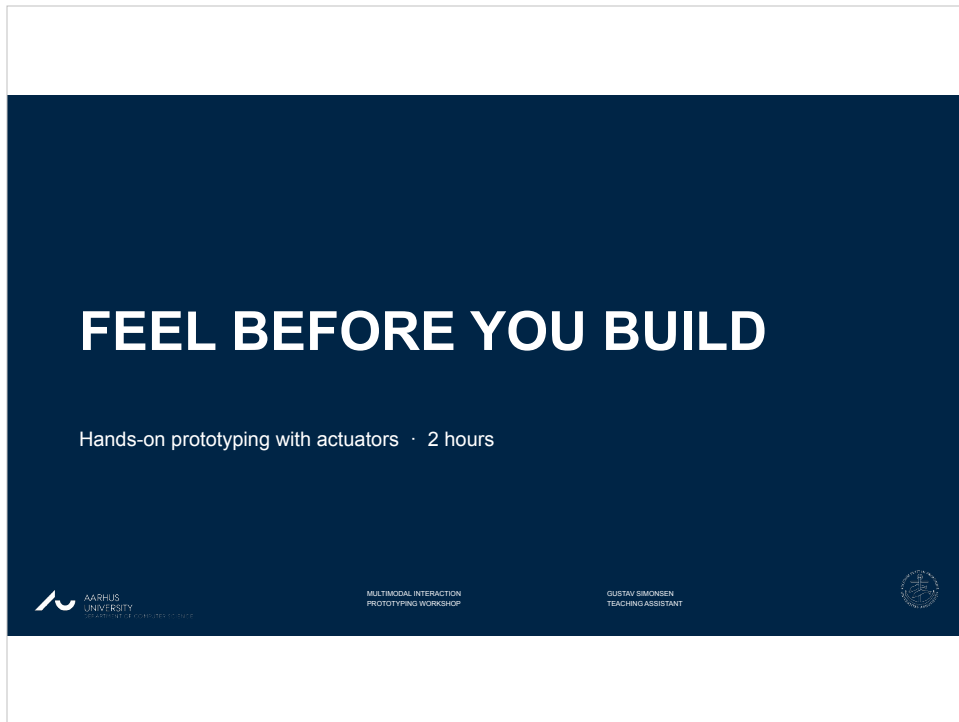
THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop



0:00. Start here. Don't read the slide.

Open with the failing demo, not with talking.
Have the coin motor and the solenoid both wired and ready before anyone sits down.

Buzz a phone-style notification on the ERM.
Then send the same message on the solenoid. Ask: which felt more urgent? Let them answer. Nobody needs theory for that, and that IS the argument for the whole session.

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46



Say the full version out loud, the slide only carries the skeleton:

"You can read a datasheet. You have never held one of these. Today you put a vibration motor against your own wrist, decide it feels cheap, and pick something else. Better now than in week 46 with the demo three days out."

Every year projects end up on the screen and the speaker, and touch gets designed on paper then dropped.

Not because it was wrong. Because nobody had held the parts

HOW TODAY WORKS

Six benches, already wired and running

- Ten minutes each. Hard timer
- Change one parameter, answer one question
- You wire nothing today

Then build one signal

- Someone else has to read it



Set the rules firmly, because the rotation collapses if you don't.

Nobody wires anything today. Everything is pre-wired and running. They turn one knob and answer one question.

Say out loud: the timer is hard. Ten minutes means ten minutes. Bench 3, the solenoid, is the one people will not want to leave.

RUN SHEET

Times are offsets from the start. The rotation is the spine of the session, so keep it moving.

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Show it, don't narrate it. Fifteen seconds.

Practical: assign each group a DIFFERENT starting bench so they rotate cleanly. Six groups, six benches. Write starting numbers on the whiteboard.

If a bench breaks mid-session, pull it and go to five. Five working benches beat six with a mystery.



0:10. Tour starts. Five minutes a bench, not ten. Set a visible timer.

You are now a timekeeper, not a lecturer. Call the rotation loudly every ten minutes. Circulate, but do not rescue. Let them poke at things.

The bench slides that follow are reference for YOU and for anyone who wants to look back. Do not present them one by one before the rotation. That would eat the hour.

01 ERM COIN MOTOR

VIBRATION

An off-centre weight on a motor shaft.
The same part as in your phone.

Speed and strength are welded together.
"Gentle but fast" is not on offer.

CARD QUESTION

At what point does it stop feeling like information and start feeling like a malfunction?

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UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT



USED FOR

phone notifications, game controller rumble,
anything worn under clothing.

part 10mm 3V coin ERM

drive 2N2222 + 1N4148

pin D9 (PWM)

draw ~75 mA

spin-up 20-40 ms

cost ~12 kr

The core limitation, made physical: speed and strength are mechanically linked on this motor. You cannot get gentle-but-fast.

Common question: "why does it feel weak at low PWM?" Below about 90 the motor cannot overcome its own friction. That floor is real and worth naming.

Never let anyone rewire this to a bare pin. 40 mA limit, motor pulls 75.

02 LRA + PIEZO DISC

VIBRATION

Two ways out of bench 01's compromise.

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CARD QUESTION

Which of the three vibrators would you actually put on a wrist, and why not the others?



The escape from bench 1's compromise. LRA is the phone-keyboard feel: 5 ms on, 5 ms off. Piezo is a crystal that flexes when you put current across it, so it is sharp but shallow.

If the LRA sounds wrong, the DRV2605L is in ERM mode. It must be told which type it's driving. That's the failure to expect here.

Good moment to point out: 123 built-in waveforms means they don't have to design vibration from scratch in their project.

03 SOLENOID TAP

IMPACT

One 15 ms pulse against a fingertip.

A knock reads as a person tapping you.

Buzzing reads as a machine.

CARD QUESTION

How many taps before it goes from alert to nagging?

USED FOR

navigation cues on the body, braille cells,
alerts that must cut through noise.

part 5V push-pull

drive MOSFET + flyback

pin D6

peak ~1.1 A

duty <= 25%, gets hot

cost ~45 kr



The memorable one. A single knock reads as a person tapping you rather than a machine signalling. Worth naming out loud, because it is the whole reason it feels different, and the slide only says the short version.

Watch the duty cycle. Solenoids get hot fast, so if it has been hammering for ten minutes, let it rest.

Separate supply, common ground. On USB alone the board browns out and resets, which looks like a code bug and isn't.

04 CAPACITIVE TOUCH

TOUCH INPUT

Bare finger on a surface. Servos react.

No button. No sensor you can point at.

The input is a wire behind a plate.

CARD QUESTION

Put your hand near the plate without touching. When exactly did it decide that was a touch?

USED FOR

invisible controls in wood or fabric, waterproof panels, touch-sensitive prototypes.

sense ADCTouch on A0
baseline 25-sample roll
trigger 1.01x average
debounce 100 ms
servos D4 + D13
cost ~0 kr, a wire



The surprise bench. No breakout board. The input is a wire.

Have the Serial Plotter on a screen here. Watching the baseline wander while the trigger line tracks it teaches the whole idea in ten seconds.

Let someone hold their hand NEAR without touching. The gradual approach is what makes the threshold question real.

BENCH 04: THE WHOLE TRICK

Touch input for the price of a wire.

`reading > baselineAverage() * 1.01`

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched



• Or a long press teaches it that a finger is **normal**

ALL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT



Slow down here. This is the one idea most worth stealing from today.

Two things students get wrong:

1. They use a fixed threshold. Works in the morning, fails after lunch,
because the reading drifts with humidity and mains hum.

2. They feed the baseline buffer always. Then a long press teaches the
baseline that a finger is normal, and the touch silently stops working.

The buffer must ONLY learn while untouched

05 PELTIER WARM / COOL

THERMAL

Heats one side, cools the other.

Flip the current, it reverses.

Sit with it. Seconds before you are sure.

Hot alternating with cold reads as lukewarm.

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"The room is busy", never "you have mail".

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Thermal. People are bad at reading thermal patterns, and this is where they find that out for themselves.

Make them time themselves. The 3 to 8 second onset kills a lot of bad project ideas before they get proposed, which is exactly what you want happening now rather than in week 46.

Safety: heatsink on the hot side, capped at 45 C. Skin burns above 50.

06 THE SAME SIGNAL, TWICE

AUDIO + TOUCH

One driver. 40 Hz you feel, 400 Hz you hear.

Together is not louder. It is more certain.

Two channels, same message.

CARD QUESTION

With ear defenders on, does the signal still work?

USED FOR

noisy or bright environments, and accessibility.
If one channel fails, one lands.

part bone-conduction

amp PAM8403 class-D

pin D11 (tone)

felt ~40 Hz

heard ~400 Hz

cost ~110 kr



Redundancy, made physical. One transducer on a plate, two frequencies: 40 Hz you feel, 400 Hz you hear, both from the same driver.

The point isn't that together is louder. It's that together is more CERTAIN.
Sending the same thing down two channels is the cheapest reliability their projects can buy.

Ear defenders must actually be used or the comparison is dishonest.

NOW MAKE ONE DO SOMETHING



0:50. Tasks start. This is the half of the session they will remember having done, rather than watched.

Hand out boards. One per pair is fine, one each is better. Everything is on GitHub, and the repo link is on the next-but-one slide.

Say clearly: you do NOT have to finish all five. Getting task 1 working and then properly playing with task 2 beats rushing through all of them.

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

- 01 Make something spin
- 02 Make something tap
- 03 Make something warm
- 04 Make it notice you
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1 to 3 are output. 4 is input. 5 is both.

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MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

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TEACHING ASSISTANT



Walk the list in about ninety seconds, no more.

The arc matters and is worth naming: 1 to 3 are output, 4 is input, 5 is both together. By task 5 they have built a complete interaction loop, which is the shape of most things they will make in the project weeks.

Point out that task 5's last experiment IS the closing brief, so the work carries straight over.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

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The format is the same every time, so say it once here rather than at every bench.

CHANGE ME at the top: two or three numbers, already set to something that works. THINGS TO TRY at the bottom: four experiments, easy to hard, and the last one usually asks them to rewrite a function rather than tweak a value.

Common upload problems, in the order they will hit you:

- * wrong board or port under Tools
- * ADCTouch not installed (tasks 4 and 5)
- * solenoid bench: board resets on fire – the

INPUTS, AND PRIOR ART



0:40. Tour over. Round-the-room now.

One sentence per group: what surprised you.
No slides, no laptops. Ten minutes.

Expect the thermal group to report the delay.
Let that land, it is more
persuasive coming from a peer than from you.

INPUTS WORTH KNOWING ABOUT

One table, not a rotation. Wander over during the build.

- Capacitive touch. A wire. See bench 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all



Mention, don't teach. These sit on one table with a sign. The slide is one line each; the caveats below are what actually saves them time.

The honest warnings matter more than the specs: PPG is garbage once the hand moves, GSR drifts so only relative change is usable, FSRs need per-pad calibration.

Push the phone option hard. For a lot of projects it genuinely beats everything else on the table and needs no soldering.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes



Five minutes, and show the actual repos on screen if you can. The slide names them; you supply the detail.

The moving screen has a browser control panel over the serial port, which is exactly what you want at the project demonstrations.

The argument: this was built by students at your stage, not by a lab with a budget. The gap between "we should use haptics" and "it moves when you touch it" is smaller than it looks.

Note for you: the sock repo has a hardcoded



BUILD ONE SIGNAL



1:40. Blind test. Thirty minutes.

Groups pick any actuator from the rotation.
Most will pick bench 1 or 3
because they are simplest. That is fine, the
constraint does the work.

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are



Blind test: they name all three

- Note which two got confused, and why

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

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TEACHING ASSISTANT



Read the brief out once, then get out of the way. Full wording, since the slide is only the bones:

"Pick one actuator. Encode three messages: arrived, something wrong, finished. Vary only the rhythm of the pulses and how strong they are. Then hand it to another group with the screen turned away and see if they can name all three."

Rhythm and strength only. If someone asks to add a second actuator, say no. The constraint is the exercise. Encoding within one channel is the skill.

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.



This is the line worth repeating twice.

Wrap up. Swap groups, receiver looks away, guess all three.

Then have each group name which two got confused and what would have separated them. That confusion is the actual learning, not the successful ones.

Pack down: everything back in the box, including the resistors. It will not all come back.



Point them at the repo. Bench 01 and 04 sketches are commented for them.

Last thing to say: pick your actuator before you design the interaction, not after. Today was so they can do that from experience rather than a datasheet.

If anyone wants to borrow parts for the project weeks, now is when they ask.